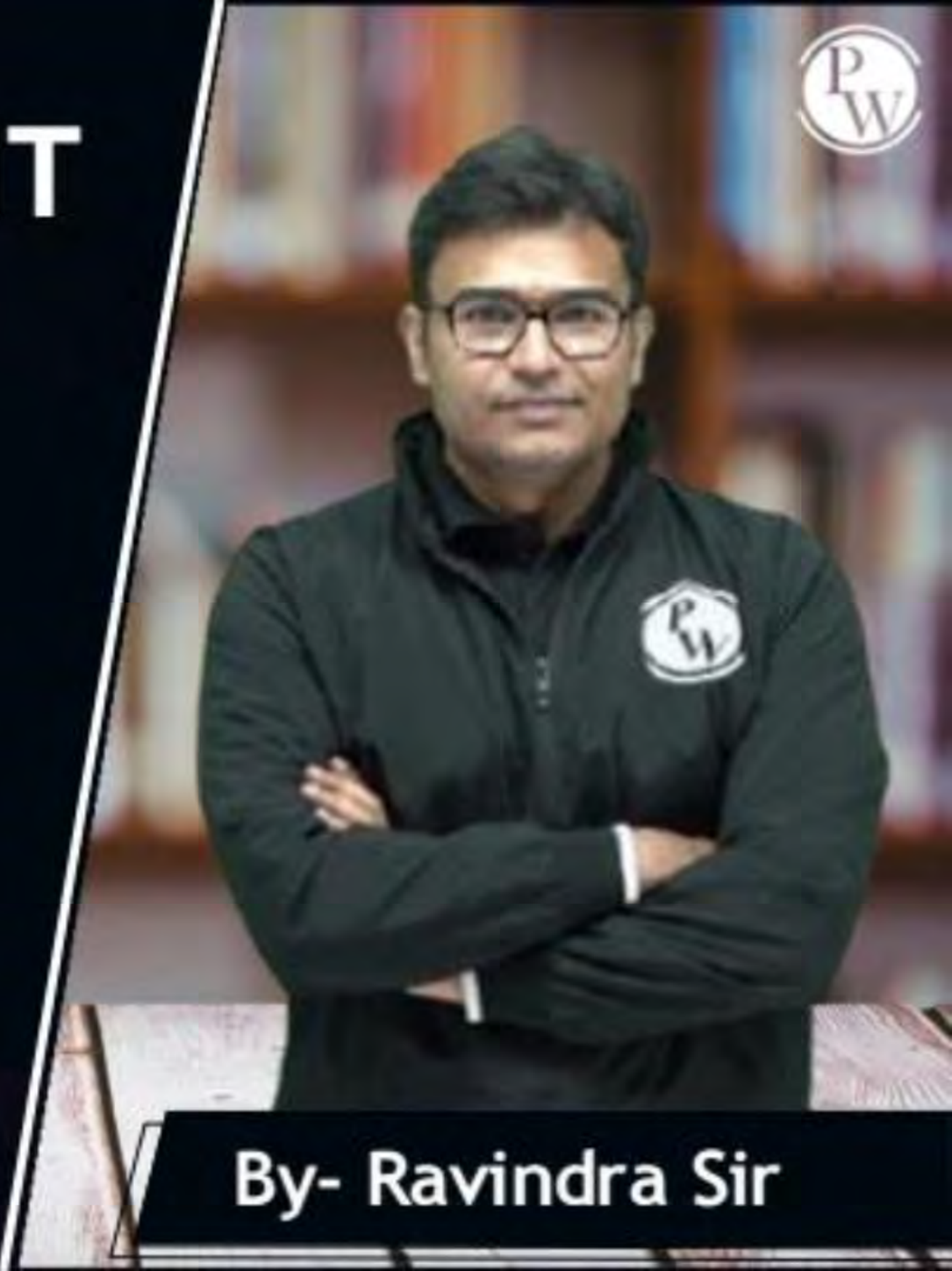
Computer Science & IT

COMPILER DESIGN
(CD)

SDT

Lecture No. 4



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

SDT

Topics to be Covered



- Topic
- Topic
- Topic

SDT

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Inspiring Stories : Mula Gabharu

Mula Gabharu – The Queen Who Raised Women Soldiers

Background: Mula Gabharu was the daughter of an Ahom king in Assam in the early 1500s.

Struggles: Her husband fell fighting an invading army. Her people were helpless and losing hope.

Achievements: She trained women warriors, led them into battle with courage, and confronted the enemy herself before being martyred.

Impact: If she could turn grief into action and lead women into battle, you can turn doubt into study and lead your rank up.



Inspiring Stories : B. N. Jagatap

B. N. Jagatap

Background: Grew up in a modest family in rural Maharashtra with no strong science background at home.

Education: PhD in Nuclear Magnetic Resonance (NMR) from IISc Bangalore.

Career Achievements: Improved NMR methods to study molecules in great detail, helping drug design and materials research.

Impact: Made high-end science tools more effective for Indian research labs and universities.



Inspiring Stories : Veeramangai Kuyili

Veeramangai Kuyili – The Woman Who Became a Human Torch

Background: Kuyili was a trusted soldier of queen Velu Nachiyar from Tamil Nadu in the 1700s.

Struggles: She worked without praise and lived in a time when women were rarely seen in battle.

Achievements: During one attack, she covered her body in ghee, set herself on fire, and jumped into the British armory. Her act became a key turning point in the war.

Impact: If she could light herself on fire to save her queen, you can light up that answer sheet with focus



Inspiring Stories : K. J. Rao

K. J. Rao

Background: Born in a small town in Karnataka with very limited means. Had to work hard to afford studies.

Education: PhD in Solid State Chemistry from IISc Bangalore.

Career Achievements: Developed new ceramic materials that can store energy and work in extreme heat.

Impact: Helped India make better energy storage systems and heat-resistant tools for industry.

SDT for type checking :-

$E \Rightarrow E_1 + E_2$ { if (E_1.type == E_2.type) && (E_1.type = int) then E.type = int else end }

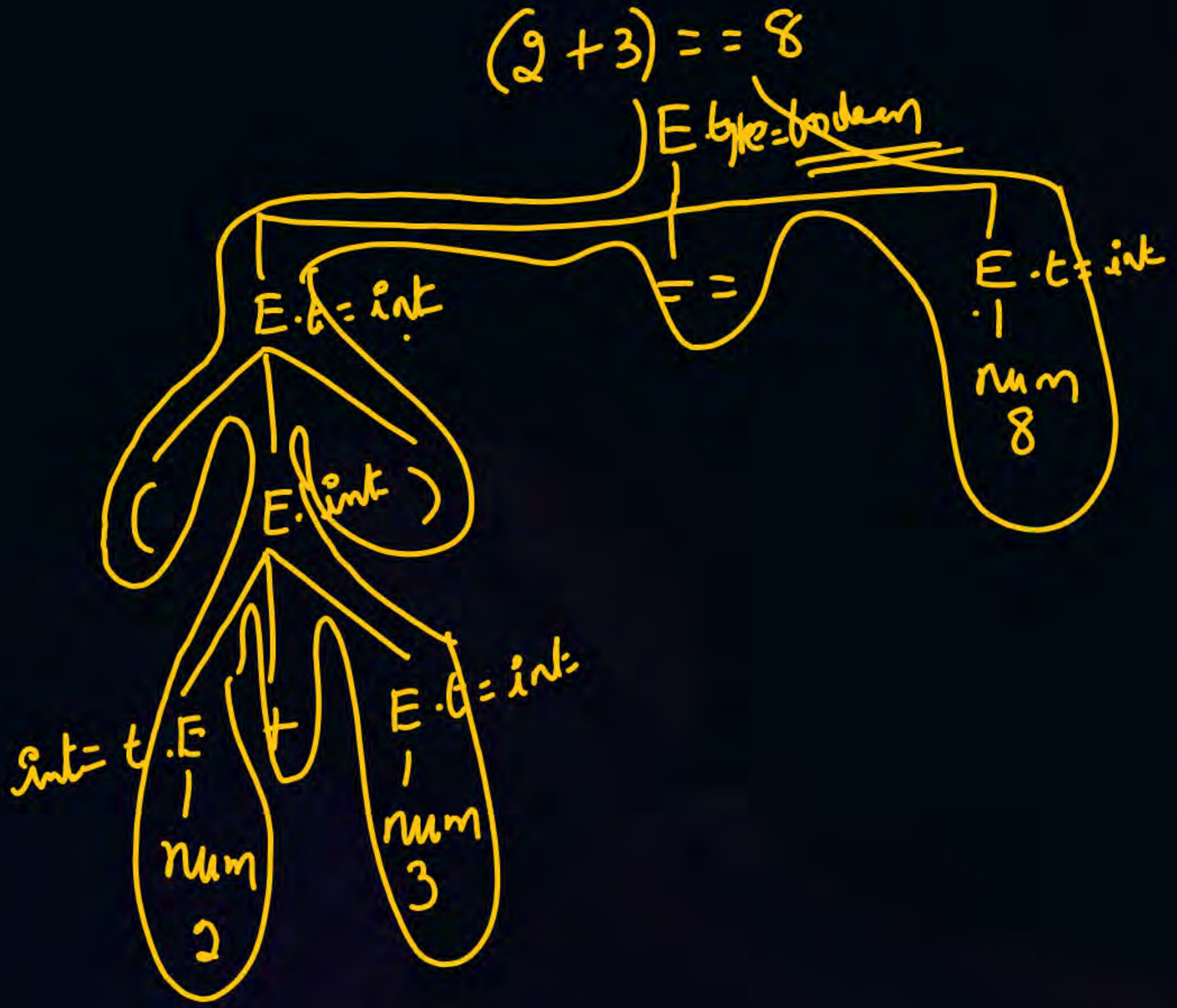
/ $E_1 == E_2$ { if(E_1.type = E_2.type) && (E_1.type = int/boolean) then E.type = boolean else end }

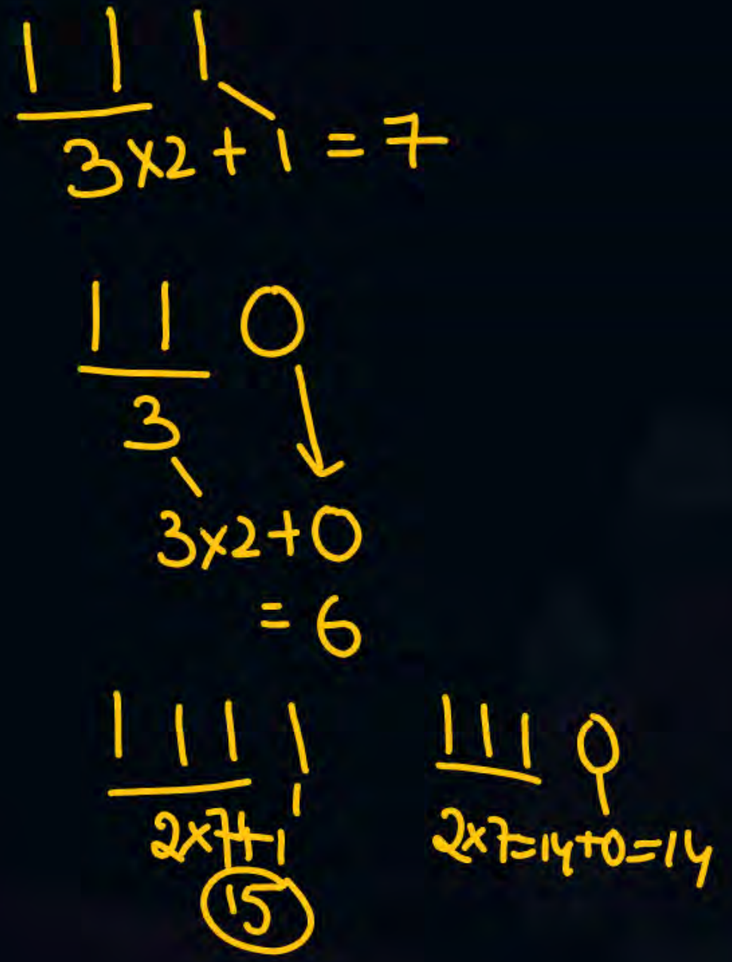
/ (E_1) { E.type = E_1.type; }

/ num { E.type = int; }

/ True { E.type = boolean }

/ False { E.type = boolean }


(2+3)==8
E type=boolean
E.t=int
==
E.t=int
num
8
E.t=int
(
E.t=int
)
int=t
E.t=int
+
E
num
2
E.t=int
num
3


1 1 1
3x2+1=7
1 1 0
3
3x2+0
=6
1 1 1 1
2x7+1
15
1 1 1 0
2x7=14+0=14

0.1 (2^{-1})

0.5

0.11 (2^{-1} 2^{-2})

0.75

0.1

$\frac{1}{2^1} = 0.5$

0.11

$\frac{3}{2^2} = 3/4 = 0.75$

0.001

$\frac{1}{2^3} = 2^{-3}$

	Count 1's	Count 0's	Count bits	Converting binary to decimal
				$N.dval = L.dval$
$N \rightarrow L$	$\{N.c = L.c\}$			
$L \rightarrow L_1 B$	$\{L.c = L_1.c + B.c\}$			$L.dval = 2 * L_1.dval + B.dval$
$/ B$	$\{L.c = B.c\}$			$L.dval = B.val$
$B \rightarrow 0$	$\{B.c = 0\}$	$\{B.c = 1\}$	$\{B.c = 1\}$	$B.dval = 0$
$B \rightarrow 1$	$\{B.c = 1\}$	$\{B.c = 1\}$	$\{B.c = 1\}$	$B.dval = 1$

$N \rightarrow L_1 . L_2 \{ N.dval = L_1.dval + \frac{L_2.dval}{2^{L_2.count}} \}$

$L \rightarrow L_1 B \{ L.count = L_1.count + B.count , L.dval = 2 * L_1.dval + B.dval \}$

$/B \{ L.dval = B.dval \}, L.c = B.c \}$

$B \rightarrow 0 \{ B.c = 1, B.dval = 0 \}$

$/1 \{ B.c = 1 ; B.dval = 1 \}$

SDT to generate 3-add code.-

$S \rightarrow id = E$ { gen(id.name = E.place) ;

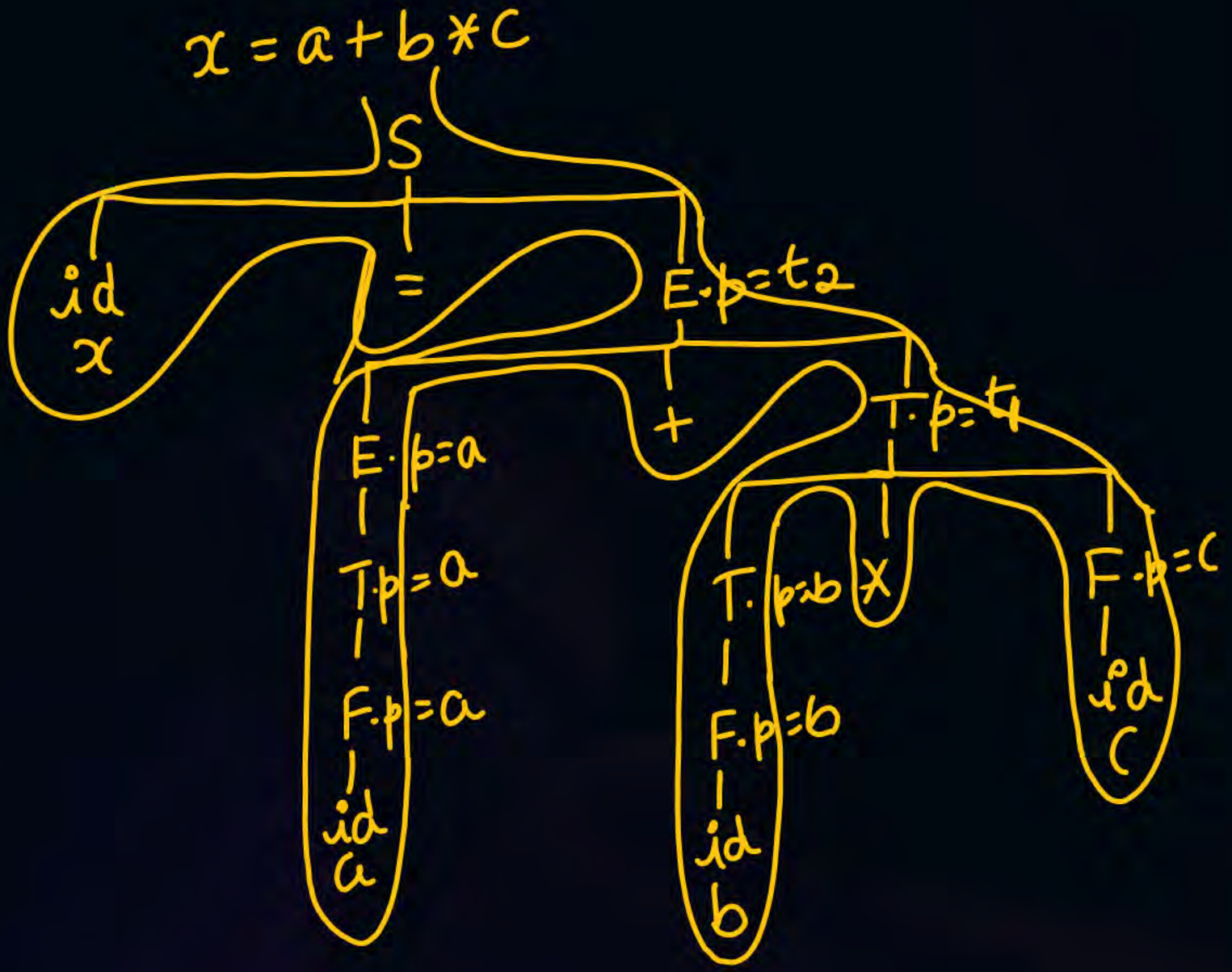
$E \rightarrow E_1 + T$ { E.place = new temp()
gen(E.place = E_1.place + T.place) }

/T { E.place = T.place }

$T \rightarrow T_1 * F$ { T.place = new temp() ;
gen(T.place = T_1.place * F.place) }

$F \rightarrow id$ { F.place = id.name; }

$T \rightarrow F$ { T.place = F.place }



$t_1 = b * c$

$t_2 = a + t_1$

$x = t_2$

→ Consider the following SDT. If an LR parser carries out translations on an i/p string "aabbccdb" what is the output?

$S \rightarrow aaS$ { print "a" }
/ bA { print 'b' }
/ b { print 'c' }

$A \rightarrow bcA$ { Print 'd' }
/ cdS { print 'e' }



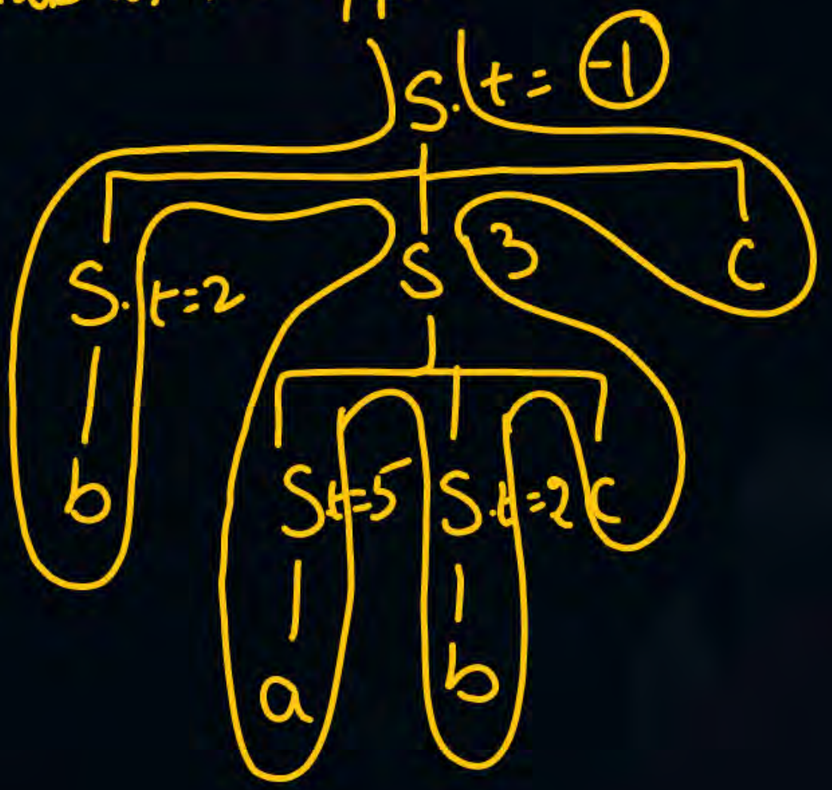
"ccdba"

→ Consider the following SDT. If an LR parser carries out the translation on an i/p string babcc. what is the o/p

$S \rightarrow S_1 S_2 C \quad \{S.t = S_1.t - S_2.t\}$

/a $\{S.t = 5\}$

/b $\{S.t = 2\}$



THANK - YOU